

Applied Data Science Capstone Project.

Space X booster analysis

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OUTLINE



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EXECUTIVE SUMMARY



- The outcome of a rocket launch is predictable based on key variables to an accuracy of 94% and using a Decision Tree Model.
- Given the Launch Site, Booster Version, Payload Mass, and Landing Site we could estimate the cost of a rocket launch for Space X and bid accordingly for Space Y to be competitive.

INTRODUCTION



- Space Y's objective is to estimate the price of each rocket launch conducted by Space X.
- The rocket's First Stage can often be reused, decreasing a rocket cost.
- Using Machine Learning Model we predicted if the First Stage will land successfully.
- Based on the model's predictions we can determine the cost of a launch and bid against SpaceX.

METHODOLOGY – Data Collection & Wrangling



- We collected data, cleaned, sorted and prepared data gathered via the SpaceX REST API and information publishes on Wikipedia.
- Data was cleaned and filtered to include only the information relevant to this project.
- Missing data was replaced by the means of the other values.

METHODOLOGY – EDA and Visual Analytics



- Using SQL we did an Exploratory Data Analysis of the landing outcomes based on Payload Mass, the Launch Sites, the Booster Version used, and Landing Sites.
- By correlating attributes we can predict successful landings under specific conditions.
- Using graphs were illustration to correlation between variables.
- Using feature engineering we calculated how important each variable is.
- Interactive visualization was used to continue the analysis and present results

METHODOLOGY – Predictive Analysis



- Build a machine learning pipeline to predict landing success.
 - Data will be preprocessed and split
- Different models will be training using the split data.
 - Logical Regression
 - Support Vector Machine
 - Decision Tree Classifier
 - K-nearest Neighbors
- Determine the best model using:
 - Confusion matrix
 - Model accuracy

RESULTS – EDA with SQL results

Between 2010-06-04 and 2017-03-20 the Landing outcomes are listed in the table below

- Indicating that the landing location has an effect on the success rate of landings.

The maximum payload (15,600kg) can be reached by the following Booster Version

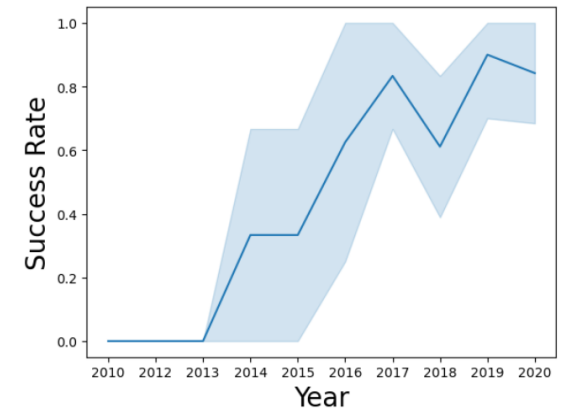
Booster_Version	PAYLOAD_MASS_KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

Landing_Outcome	count(Landing_Outcome)
Controlled (ocean)	3
Failure (drone ship)	5
Failure (parachute)	2
No attempt	10
Precluded (drone ship)	1
Success (drone ship)	5
Success (ground pad)	3
Uncontrolled (ocean)	2

RESULTS – EDA and Visual Analytics

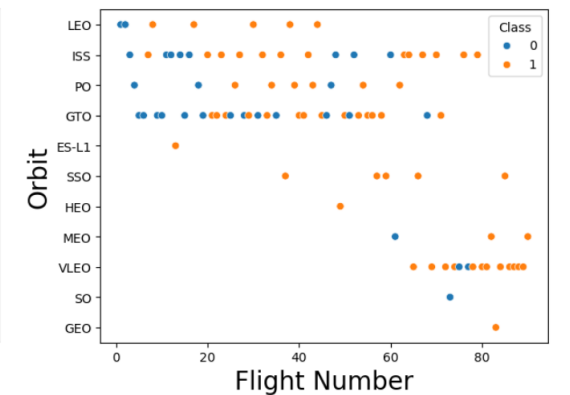
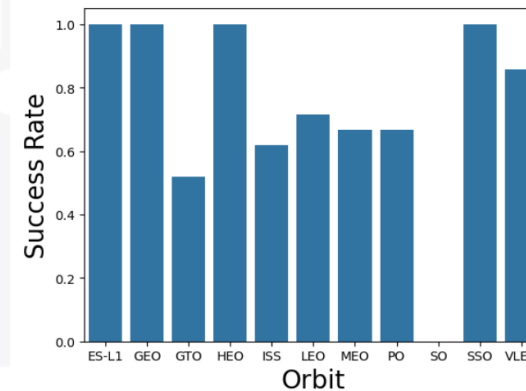
The success rate has increase between 2013 and 2020.

- Indicating that as the success rate increases with the number of flights, showing that each flight can bring valuable insight.



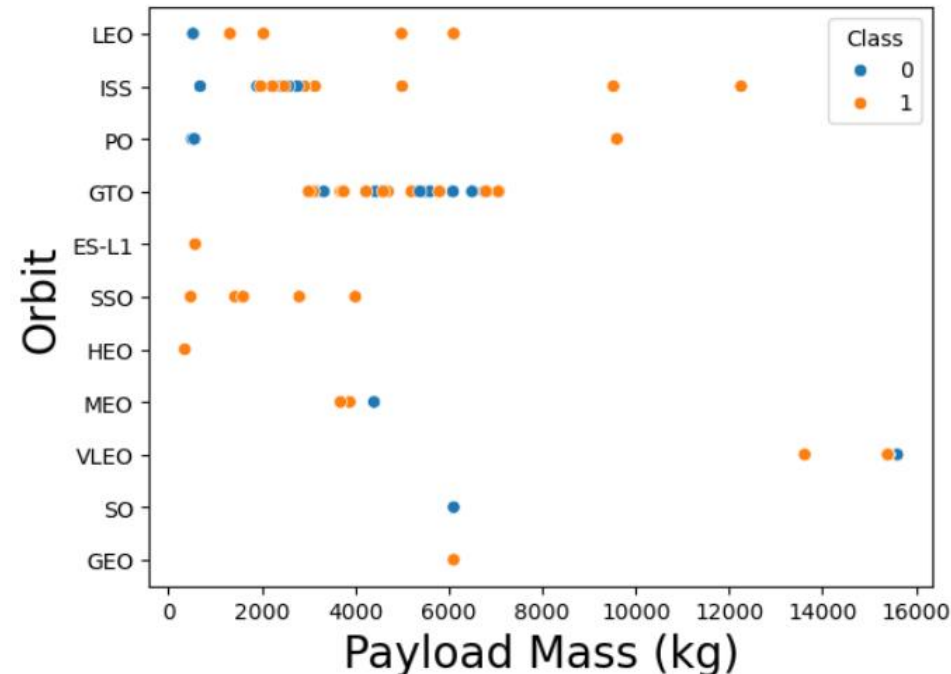
The destination orbits have different success rates.

- As flight number increases, the success rate does too.
- Supporting the first graph's conclusion above.

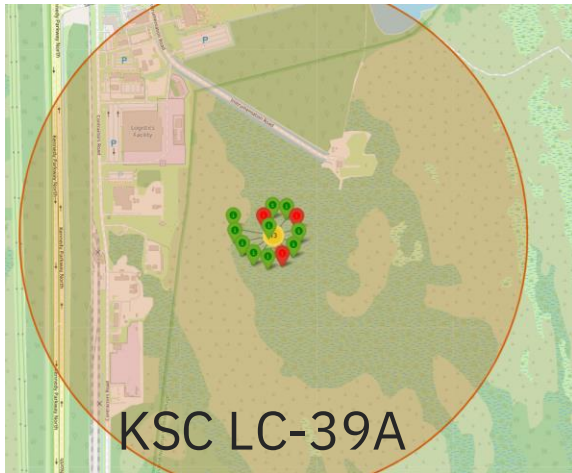


RESULTS – EDA and Visual Analytics

By combining Orbit and Payload Mass, we are able to see that the successful landing rate for Orbits Polar, LEO, and ISS is better with heavier Payload Mass.

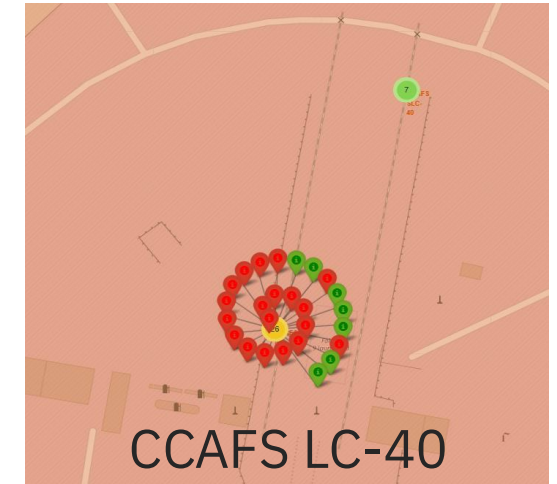
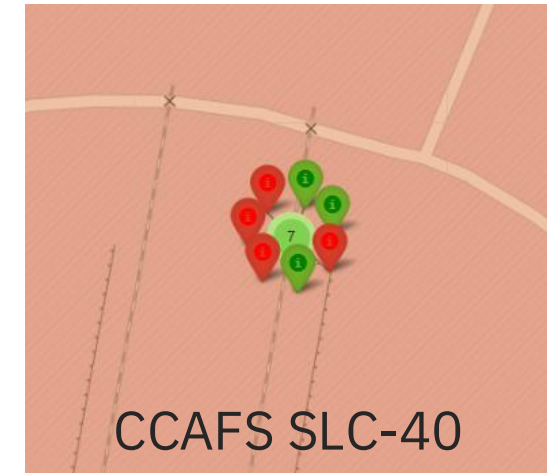
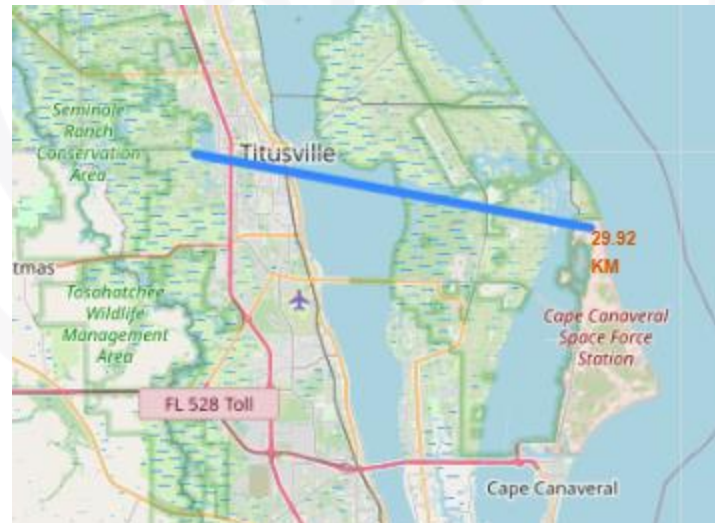
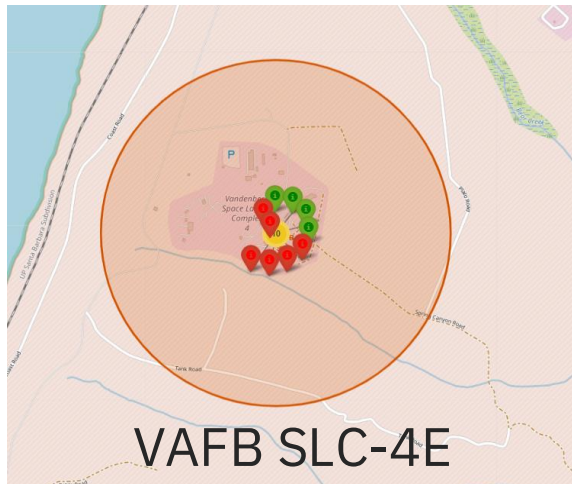


RESULTS – Folium Map results



Successful landings based on launch site.
Success in green.
Failure in red

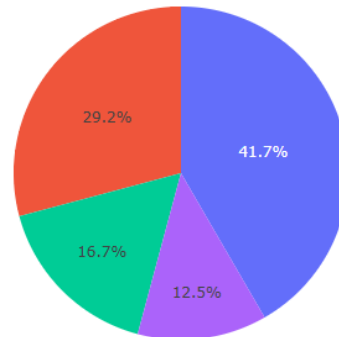
You can also use Folium maps to calculate the
distance between launch sites and geographical
landmark (like a city)



RESULTS – Dashboard Results

Using DASH we can easily see which launch site have the biggest success rate.
KSC LC-39A has the best rate

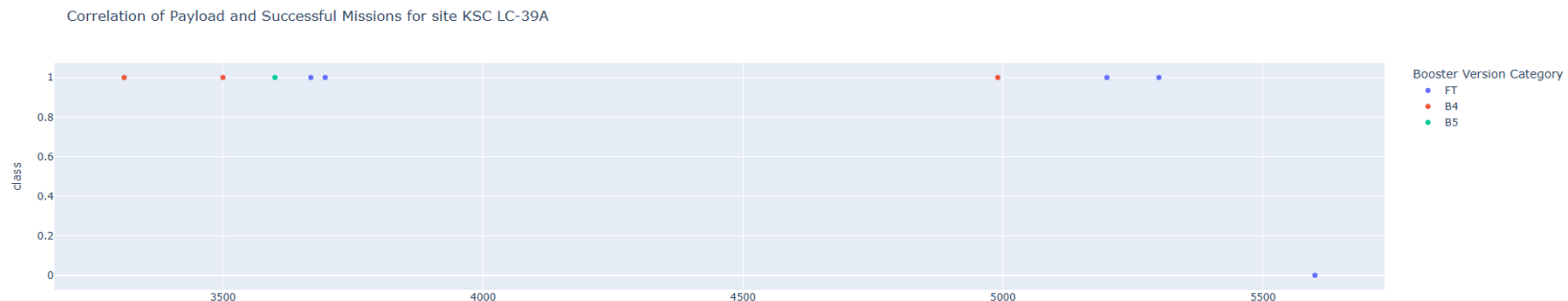
Success Count for ALL Launch Sites



■ KSC LC-39A
■ CCAFS LC-40
■ VAFB SLC-4E
■ CCAFS SLC-40

RESULTS – Dashboard Results

We can investigate further the relationship between BoosterVersion and the Payload for each launch site.

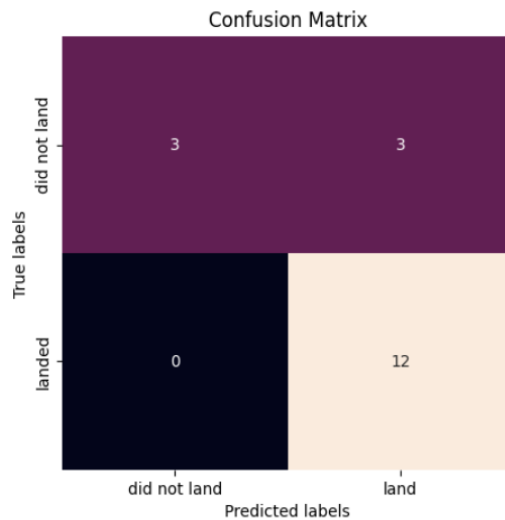


RESULTS – Predictive Analysis - Accuracy

Model used on split data	Accuracy of test data
Logical Regression (LR)	83.33%
Support Vector Machine (SVM)	84.82%
Decision Tree Classifier (TREE)	94.4%
K-nearest Neighbors (KNN)	83.33%

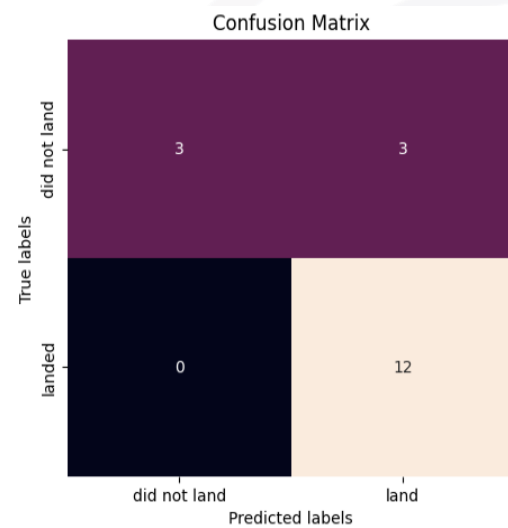
RESULTS – Predictive Analysis – Confusion Matrix

LR



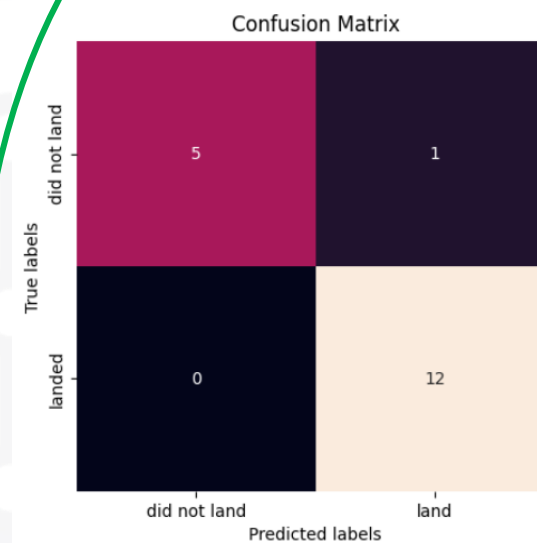
True +	False -
12	3
Accuracy 83.33%	

SVM



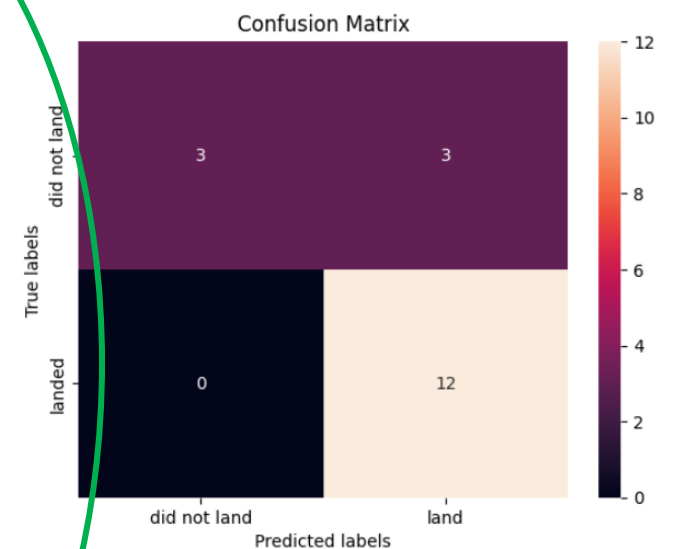
True +	False -
12	3
Accuracy 84.82%	

TREE



True +	False -
12	1
Accuracy 84.82%	

KNN



True +	False -
12	3
Accuracy 83.33%	

CONCLUSION



The decision tree model is the best way to predict the success of landing the First Stage, and therefore estimate the price of a launch based on the Booster Version, Payload Mass, Launch and Landing locations.